POS DGFY Location Switching Implementation Plan

Editable consolidated plan | Plan v1.0 | 27 September 2026

Project	POS-DGFY - Global Location Switching
Plan version	1.0
Revision date	2026-09-27
Plan status	NOT EXECUTION READY
Planning depth	Level 4 - critical multi-tenant, financial, inventory, authorization, and migration change

Purpose. This editable document consolidates the complete Location Switching planning package into one Word document. It defines the recommended global location context, the existing Shift-controlled transition, module behavior, permissions, data ownership, verification, rollout, recovery, and execution handoff.

Recommended approach

Use a visible Current Location entry point in the global header and mobile sidebar. Route operating-location changes through the existing Shift workflow, and keep viewing-location access read-only. Do not implement a second unrestricted switching path.

Plan contents

- PLAN-01 Executive Summary
- PLAN-02 Problem and Current State
- PLAN-03 Target Architecture and Data Model
- PLAN-04 Scope Requirements and Business Rules
- PLAN-05 Backend Permissions and Security
- PLAN-06 Frontend UX and Cross Module Behavior
- PLAN-07 Migration and Backward Compatibility
- PLAN-08 Testing Verification and Acceptance
- PLAN-09 Rollout Observability and Recovery
- PLAN-10 Risks Dependencies and Architecture Decisions
- PLAN-11 Final Recommendation and Execution Handoff

PLAN-01 Executive Summary

Objective

Design a secure global Location Switching capability for the POS so changing the active branch updates every location-dependent
module without mixing inventory, sales, cash, permissions, reports, settings, vouchers, price lists, affiliates, or audit history.

Recommendation

Add a visible Current Location entry point in the global header/sidebar, but keep the existing Shift workflow as the only authority for
changing the operating location. Maintain separate operating-location and viewing-location contexts.

Why this fits the existing system

- The backend already exposes pos:switch_location and POST /terminal/shifts/:id/switch-location.
- TerminalPage already tracks operatingLocationId, viewingLocationId, and queueLocationScopeId.
- The Shift UI already validates destination terminals, cash values, current carts, and offline operations.
- ADR 0009 and ADR 0031 already require location-scoped inventory, terminal binding, shift-safe transitions, and auditability.

External research

Square uses an explicit in-app Switch location action, location-specific reporting, inventory, devices, and permissions, and warns
against switching with pending offline payments. Toast exposes a location selector while separating location access from
per-location permissions.
Sources: https://squareup.com/help/us/en/article/5580-manage-multiple-locations-with-square ;
https://squareup.com/us/en/point-of-sale/features/multi-store ;
https://support.toasttab.com/en/article/Assigning-Employees-to-Multiple-Locations ;
https://support.toasttab.com/en/article/MLM-Permissions-Guide

Plan status

NOT EXECUTION READY: Phase 354 discovery is still required for voucher, price-list, affiliate, and setting scope, and the
phase ledger must be reconciled with the existing Phase 354-357 location-access documents. The execution skill must
not begin implementation until these gates are closed.

Primary acceptance outcome

An authorized user can move from one authorized operating location to another through an atomic, audited, shift-safe flow, while
read-only users can view authorized location data without gaining transaction or mutation rights.

PLAN-02 Problem and Current State

Problem statement

The POS needs one understandable location context, but operating a branch and merely viewing branch data have different safety
requirements. A free global switch would risk cross-branch sales, stock, cash, queue, and audit errors; a Shift-only entry point is
difficult to discover.

Confirmed current state

Area
Evidence
Classification
Permission
apps/dgfy-api/src/config/permissions.js:115 defines pos:switch_location.
REUSE
Operating transition
apps/dgfy-api/src/routes/pos.js:343 exposes the governed shift switch endpoint.
REUSE
Frontend service
packages/web-core/src/features/pos/services/posService.js:686 posts the transition.
REUSE
Context state
TerminalPage.jsx:720-722 tracks operating, viewing, and queue scopes.
EXTEND
Admin viewing
TerminalOperationsWorkspace.jsx:768-839 exposes Register Branch and Viewing Branch
controls.
EXTEND
Shift UI
TerminalOperationsWorkspace.jsx:1131+ contains Change Active Shift Location.
REUSE
Inventory authority
ADR 0009 identifies item_location_stocks and location movements as stock truth.
LEAVE UNCHANGED

Current checkout baseline

- Branch: feat/DGFY-343-2067-pos-workflow-consolidation.
- Baseline commit observed during planning: 786cd4e8329db1a65398ce6aa7eb020ffc8f3b28.
- The working tree is dirty with unrelated application, generated, and local skill changes. Implementation must preserve them and

must use a clean, explicitly selected worktree or branch.

Root cause and secondary effects

The root cause is not the absence of location logic. The system already has partial location context and a governed shift transition.
The gap is that the context is not yet presented and reconciled as one cross-module product contract, and not every requested
module has a confirmed scope contract.

- Secondary effect: users may interpret viewing a branch as operating that branch.
- Secondary effect: adding a global selector without request invalidation could display stale data from the previous branch.
- Secondary effect: modules with unclear scope could either leak data or unnecessarily block valid operations.

Documentation constraint

The main implementation phase ledger currently has a high-water mark below the location-access documents, while those
documents define Phase 354 and runtime Phases 355-357. Reconcile the ledger before implementation; do not invent a new
phase sequence.

Stakeholders

Stakeholder
Need
Risk
Cashier
Fast, obvious branch context and safe switch flow.
Accidental sale or cash assignment to wrong branch.
Branch manager
Location-scoped operations, staff, stock, reports, and
settings.
Unauthorized cross-location mutation.
Owner / administrator
Authorized cross-location viewing and controlled
administration.
Overbroad access and confusing operating state.
Finance / audit
Stable historical reports and traceable transitions.
Rewritten history or missing actor attribution.
Engineering / operations
Observable, reversible rollout and clear recovery.
Partial transitions and difficult incident repair.

PLAN-03 Target Architecture and Data Model

Target context model

LocationContext = {
tenantId,
operatingLocationId,
viewingLocationId,
terminalId,
activeShiftId,
queueLocationScopeId,
contextVersion,
transitionState
}
The frontend may display and coordinate this context, but the backend remains authoritative for permissions, terminal binding, shift
ownership, location grants, and transaction location.

Context semantics

Context
Controls
Allowed changes
Operating location
Checkout, stock deductions, incoming queue, parked-sale
payment/resume, cash drawer, shift.
Only through governed shift transition or valid shift opening.
Viewing location
Read-only reports, audit, monitoring, historical data.
Can change without changing register or shift when permission
allows.
Queue scope
Incoming order visibility and operational refresh.
Derived from active operating shift; not independently
user-controlled.

Source of truth

Concept
Authoritative owner
Derived or historical behavior
Location access
Tenant user/location grants.
Re-evaluate at every protected request.
Terminal location
Registered terminal record.
Strict terminal mode binds the device to its registered location.
Operating location
Active shift.
Cannot be changed by a client-only state update.
Stock
item_location_stocks and inventory movements.

items.current_stock remains derived aggregate where applicable.
Catalog identity
Catalog item, SKU, category, and base-price records.
Do not create branch copies of the same item.
Completed sale
Immutable transaction location snapshot.
Later context changes do not rewrite it.
Transition
Atomic shift-location transition record.
Records source, destination, actor, cash, reason, and outcome.
Concept
Authoritative owner
Derived or historical behavior
Audit
Append-only tenant audit trail.
Filter by location for viewing; do not create separate histories.

ADR-001 - Global context with governed operating transition

Decision: use a global header/sidebar entry point that dispatches to the existing Shift-owned operating transition, while preserving a
separate read-only viewing context.
Alternatives rejected: Shift-only switch for poor discoverability; unrestricted sidebar switch for safety risk; duplicate catalogs for data
drift; separate audit stores for incomplete investigation.
Consequences: more explicit context labels and refresh coordination are required, but the design reuses current contracts and is
reversible through a feature flag.

ADR-002 - Shared catalog, location-specific operational state

Decision: keep item identity, SKU, category, and tenant-wide base price centralized; keep stock, availability, visibility, and any
approved overrides location-aware. Add schema only when discovery proves a required location dimension is missing.

Target state diagram

User authentication
|
v
Permission + location grant checks
|
+--> Viewing context ----> Reports / Audit / Monitoring (read-only)
|
+--> Operating context --> Terminal + Shift --> Sales / Stock / Queue

PLAN-04 Scope Requirements and Business Rules

In scope

- Global location entry point and context labels.
- Operating-location switching through the existing Shift workflow.
- Location-aware Items, stock, Add Item, sales, parked sales, queue, reports, shifts, settings, vouchers, price lists, affiliates, and

audit.

- Server-side permissions, concurrency, idempotency, offline safety, testing, rollout, observability, and recovery.

Out of scope

- GPS, Wi-Fi, IP, or automatic physical branch detection.
- Automatic conflict takeover or silent shift reassignment.
- Duplicate branch catalogs or destructive schema replacement.
- New offline login or branch transfer behavior not already supported by the contract.

Functional requirements

ID
Requirement
Priority
Verification

- FR-001

Authorized users can open a global Current Location entry point.
P1
Desktop/mobile UI test

- FR-002

The UI distinguishes operating, viewing, and queue context.
P1
UI and E2E test

- FR-003

An active operating switch uses the governed close/open shift flow.
P0
API integration test

- FR-004

Only authorized destination locations and terminals are selectable.
P0
Authorization test

- FR-005

Sales, stock, queue, parked sales, and payments use the operating location.
P0
Cross-module E2E

- FR-006

Reports and audit support authorized viewing-location filters.
P1
Report/API test

- FR-007

Settings, vouchers, price lists, and affiliates apply explicit scope rules.
P1
Module contract tests

- FR-008

Every successful or rejected transition is traceable.
P0
Audit verification

Quality, security, data, and operations requirements

ID
Requirement
Priority

- NFR-001

No stale request from the old context may overwrite the new context.
P0

- NFR-002

Switch submissions are idempotent and safe to retry.
P0

- SEC-001

Backend rejects forged tenant, location, terminal, shift, cashier, and operator identifiers.
P0

- SEC-002

Location permission is independent from action permission.
P0

- DATA-001

Every completed sale has exactly one immutable location snapshot.
P0

- DATA-002

Stock truth remains location-scoped and reconciliable.
P0

- OPS-001

Transition, denial, conflict, and reconciliation signals are observable.
P1

- MIG-001

Schema changes are additive and backward compatible until verified.
P0

Business rules

ID
Rule
Failure behavior
Affected areas

- BR-001

A sale belongs to one operating location and active
shift.
Reject or preserve current state.

Sales, inventory, reports, audit

- BR-002

Switch permission does not grant destination access.
Return 403.
Authorization, UI

- BR-003

No switch with unresolved cart, payment, or offline operation.
Block with actionable reason.
Checkout, shift, offline

- BR-004

Source shift close and destination shift open are atomic.
Keep source state; retry safely.
Shift, cash, audit

- BR-005

Viewing a location never grants mutation or incoming-queue access.
Return 403 or read-only state.
Reports, monitor, queue

- BR-006

Parked sales retain origin location and shift ownership.
Require explicit handoff policy.
Parked sales, payments

- BR-007

Historical transactions do not change when current context changes.
Reconcile and block release if violated.
Reports, finance

- BR-008

Strict terminal mode requires a destination registered terminal.
Reject destination.
Terminal, shift

Acceptance criteria

- AC-001: A permitted user can switch safely and the new location appears consistently in all affected modules.
- AC-002: An unassigned location, forged location ID, or missing permission is rejected server-side.
- AC-003: Pending offline work, unresolved cart, or concurrent transition cannot corrupt the shift.
- AC-004: Stock, queue, sales, reports, vouchers, pricing, affiliates, and audit use the correct context.
- AC-005: Desktop and mobile users can identify current context and understand blocked actions.
- AC-006: Rollback preserves historical transactions and supports reconciliation.

PLAN-05 Backend Permissions and Security

Backend strategy

Reuse the existing switch-location endpoint and use case. Extend only the missing module scope checks discovered during Phase
354. Keep controllers transport-only and enforce rules in use cases and repositories according to the repository architecture
boundary.

Authentication, authorization, and data scope

Layer
Question
Required check
Authentication
Who is the caller?
Authenticated tenant user, active account, valid session/operator.
Authorization
What may the caller do?
pos:switch_location plus module-specific action permission.
Data scope
Which records may the caller access?
Location grant, terminal assignment, shift ownership, and requested viewing scope.

API behavior

- Reuse POST /pos/terminal/shifts/:id/switch-location.
- Validate shift ID, terminal pairing, destination location, destination terminal, closing cash, opening float, reason, notes, and

idempotency key.

- Derive tenant, actor, cashier, source location, and current shift from trusted session and database state.
- Reject client attempts to override protected identity or scope fields.
- Return deterministic errors: 403 for authorization, 409 for conflict/state race, 422 for invalid input, and 5xx only for unexpected

failures.

Atomicity and idempotency

- Lock the source shift and destination terminal/shift records as required by the existing data model.
- Perform close/open transition and audit write in one transaction.
- Reuse idempotency keys so retries do not create duplicate transitions or cash events.
- Preserve the source state if any destination validation or write fails.

Security checks

ID
Attack or failure
Verification

- SEC-001

Forged location_id or terminal_id.
Direct API request with altered identifiers must fail.

- SEC-002

Permission without location grant.
User with pos:switch_location but no destination grant receives 403.

- SEC-003

Viewing context used for mutation.
Mutation request using viewing scope is rejected.

- SEC-004

Cross-tenant identifier substitution.
Tenant ownership query rejects foreign IDs.

- SEC-005

Replay or duplicate submit.
Same idempotency key returns the original outcome.

- SEC-006

Sensitive audit/log leakage.
Logs contain IDs and reasons, not passwords, tokens, or payment secrets.

Backend exit criteria

- All protected APIs use trusted scope checks.
- Concurrency tests prove no double-open or cross-location shift state.
- Audit and reconciliation records are produced for success and failure.
- No new API duplicates the existing governed switch flow.

PLAN-06 Frontend UX and Cross Module Behavior

Navigation decision

Place Current Location in the global header on desktop and in the mobile sidebar/menu. Clicking it opens the governed flow. Keep
the detailed Change Active Shift Location form inside Shift as the canonical workflow, not as a second implementation.

Context display

UI element
Display
Safety behavior
Global header
Operating location, terminal, and shift status.
Never imply that viewing a branch changes the register.
Location selector
Authorized destinations and read-only viewing choices.
Hide unauthorized options; still enforce on server.
Shift transition modal
Destination, terminal, closing cash, opening float, reason, notes.
Require preflight completion before submit.
Read-only banner
Viewing location and reason it is read-only.
Disable mutation and queue actions.
Loading state
Context transition in progress.
Block duplicate actions and stale navigation.
Error state
Permission, conflict, offline, or validation reason.
Offer recovery action without auto-takeover.

Operating switch workflow

Open Current Location
-> Run cart, payment, offline, permission, and terminal preflight
-> Select authorized destination location and terminal
-> Enter counted closing cash, opening float, reason, and notes
-> Submit idempotent shift transition
-> Refresh context and all dependent modules
-> Confirm destination operating context

Module behavior

Module
Operating location
Viewing location
Items / Add Item
Load branch stock, availability, visibility, and permitted branch operations.
Allow only authorized read/edit behavior defined by module permissions.

Sales / checkout
Use active shift, terminal, stock, parked sales, and queue.
No checkout or payment mutation.
Reports
Default to operating branch where relevant.
Allow authorized branch or multi-branch read-only filters.
Shifts
Owns transition, cash drawer, and terminal state.
Read-only monitoring only.
Settings
Load branch and terminal settings while preserving tenant
settings.
Allow only permitted configuration viewing/editing.
Module
Operating location
Viewing location
Vouchers
Apply location eligibility and redemption location.
View definitions only if authorized.
Price lists
Load location assignment or override.
View permitted price configuration.
Affiliates
Record attribution and eligibility location.
Filter reporting by authorized location.
Audit
Write current operating context.
Filter the tenant-wide trail by authorized location.

Refresh and race handling

- Increment a context version before transition and abort or ignore old requests.
- Clear location-dependent caches and carts only according to the approved workflow.
- Refresh items, stock, queue, reports, shift, settings, vouchers, pricing, affiliate context, and audit filters after authoritative

success.

- Do not refresh a module into a new location until the backend confirms the transition.

Responsive and accessibility requirements

- Desktop header selector and mobile sidebar selector expose the same permission-aware flow.
- Labels explicitly say Operating Location or Viewing Location.
- Focus moves into the transition dialog and returns to the trigger after close.
- Warnings are readable without color alone and controls expose disabled reasons.

PLAN-07 Migration and Backward Compatibility

Migration position

No migration is approved yet: First audit existing tables, indexes, constraints, repositories, and queries for vouchers, price lists, affiliates, settings, reports, and audit. Reuse existing location fields and transition records wherever present.

Required discovery

- Map every affected entity to tenant, location, terminal, shift, and historical location fields.
- Find all APIs accepting location_id or equivalent filters and classify whether they derive or trust it.
- Find all reports and exports that aggregate without an explicit location scope.
- Confirm whether settings are stored by tenant, branch, terminal, or mixed scope.
- Confirm voucher, price-list, and affiliate ownership and eligibility models.

Safe migration sequence

Stage
Action
Exit evidence
Prepare
Add compatible indexes/columns only when justified; keep old reads/writes working.
Migration tests and schema review pass.
Backfill
Populate location metadata from authoritative terminal, shift, transaction, or grant
sources.
Row counts and ownership parity pass.
Validate
Compare old and new aggregates, reports, and ownership.
No unexplained drift.
Switch reads
Enable location-aware reads behind a feature flag.
Pilot reads match expected data.
Switch writes
Enable location-aware writes and audit.
Integration and reconciliation tests pass.
Observe
Monitor errors, totals, stock, permissions, and audit events.
Observation window passes.
Clean up
Remove compatibility code only after explicit approval.
Rollback window closed.

Backward compatibility

- Existing terminals and current shift flows must continue working when the feature flag is off.
- Older clients must fail safely or remain on compatible APIs; do not require a new client for an existing safe path without a

migration plan.

- Historical reports must remain stable and must not be recomputed from later branch configuration.
- Generated build artifacts and unrelated dirty-worktree changes are not migration inputs.

Migration exit criteria

- Representative tenant data has been migrated in rehearsal.
- Foreign keys, unique constraints, indexes, ownership, null handling, reports, and performance are verified.
- Rollback and reconciliation are tested without deleting historical data.

PLAN-08 Testing Verification and Acceptance

Verification versus validation

Verification proves the implementation follows the approved requirements. Validation proves that users can safely operate multiple
branches without confusion or cross-location errors.

Verification matrix

ID
Requirement or rule
Method
Evidence

- TEST-001
- FR-001 / FR-002

Desktop and mobile UI
Screenshots/video showing explicit operating/viewing labels.

- TEST-002
- FR-003 / BR-004

Backend integration
Atomic close/open transition and persisted audit event.

- TEST-003
- FR-004 / SEC-001 / SEC-002

Authorization API
403 for unassigned or forged destination.

- TEST-004
- BR-003

Failure-path integration
Pending offline/cart/payment blocks switch without mutation.

- TEST-005
- FR-005 / DATA-001 /
- DATA-002

Database and E2E
Sale, stock movement, queue, and report all use one location.

- TEST-006
- FR-006 / BR-007

Report regression
Historical location remains unchanged after switch.

- TEST-007
- FR-007

Module contract tests
Settings, voucher, pricing, affiliate scope is enforced.

- TEST-008
- NFR-001 / NFR-002

Concurrency test
No stale overwrite and duplicate transition is idempotent.

- TEST-009

- MIG-001

Migration rehearsal
Counts, constraints, ownership, reports, and rollback pass.

- TEST-010
- OPS-001

Observability test
Metrics, logs, alerts, and audit evidence appear.

Required GIVEN / WHEN / THEN scenarios

- Given a user has access to Location A and B, when they switch with a clear cart and valid cash values, then the new shift and

terminal context are Location B.

- Given a user lacks Location B access, when they submit destination B directly to the API, then the server returns 403 and

Location A remains unchanged.

- Given pending offline operations, when the user selects another location, then the switch is blocked and no shift or cash state

changes.

- Given two terminals target the same destination shift, when both submit, then one succeeds and the other receives a

deterministic conflict.

- Given a sale completed at Location A, when an administrator views Location B, then the sale remains attributed to A.
- Given a parked sale belongs to Location A, when a cashier works at Location B, then it cannot be silently paid as a B sale.

Test layers

Layer
Coverage
Unit
Scope policy, context reducer, transition validation, formatting, permission decisions.
API
Authorization, validation, idempotency, conflicts, tenant and location ownership.
Database
Constraints, locks, historical snapshots, inventory movements, migration parity.
UI
Selector, modal, disabled states, labels, mobile/desktop responsive behavior.
E2E
Login -> context selection -> shift -> sale -> stock -> report -> audit -> close.
Security
Direct API bypass, parameter tampering, cross-tenant and cross-location attempts.
Recovery
Timeout, retry, refresh, restart, failed migration, rollback, reconciliation.
Performance
Large item lists, reports, location filters, concurrent users, refresh storms.

Acceptance gate

No critical requirement may be marked verified without implementation evidence and verification evidence. A passing component
test does not prove cross-module correctness.

PLAN-09 Rollout Observability and Recovery

Release strategy

Use a server-authoritative feature flag and pilot rollout for one representative multi-location tenant before broader release. The flag
must not bypass backend authorization or change historical data.

Release sequence

- Run repository checks, migration rehearsal, security tests, regression tests, and E2E evidence.
- Deploy compatible backend and migration preparation.
- Enable the feature for internal users or one pilot tenant.
- Observe transition, stock, queue, report, permission, and audit signals.
- Expand location-by-location or tenant-by-tenant only after the observation gate passes.

Success signals

Signal
Expected behavior
Switch success rate
Authorized valid switches succeed without retries or manual repair.
Authorization failures
Only expected denials; no unexpected cross-location access.
Inventory parity
Stock movements and reports reconcile by location.
Cash/shift parity
Closing/opening values and transitions reconcile.
Queue isolation
Incoming orders appear only for the active operating scope.
Audit completeness
Every transition has actor, source, destination, reason, and outcome.
UX errors
Users can understand and recover from blocked transitions.

Immediate stop conditions

- Cross-tenant or cross-location data leakage.
- Incorrect inventory movement, cash total, or financial report.
- Missing or incorrect audit attribution.
- Partial shift transition or duplicate shift opening.
- Parked sale or incoming queue appears under the wrong location.
- Critical security, migration, or API failure.

Rollback and recovery

- Disable the feature flag and restore safe application code/configuration.
- Keep additive schema changes in place until written data is reconciled.
- Reconcile transactions, stock, shifts, queue records, and audit events created during the incident.
- Preserve evidence, document root cause, update the plan and tests, and retry only after corrective verification.

Operational evidence

Capture deployment SHA, application/database version, feature-flag state, migration result, test evidence, metrics snapshot, audit
sample, and rollback readiness before declaring release success.

PLAN-10 Risks Dependencies and Architecture Decisions

Dependency graph

Phase 354 contract and discovery
-> scope matrix and permission matrix
-> Phase 355 backend enforcement
-> Phase 356 shared context and UI
-> module integration and report/audit validation
-> Phase 357 hardening
-> pilot rollout

Parallelizable work

- The module-scope inventory can run in parallel with UX wireframes after the contract boundaries are fixed.
- API authorization tests and UI selector tests can be prepared in parallel once requirement IDs are approved.
- Migration rehearsal can begin after schema ownership is confirmed, without enabling production writes.

Risk register

ID
Risk
Likelihood / impact
Mitigation and contingency

- RISK-001

Client or stale request mixes locations.
Medium / High
Context versioning, aborts, server scope checks; disable feature if leakage appears.

- RISK-002

Pending offline work is lost or assigned incorrectly.
Medium / High
Fail closed before switch; test reconnect and recovery.

- RISK-003

Voucher/price/affiliate scope is guessed.
High / High
Phase 354 schema and route audit; keep plan NOT EXECUTION READY.

- RISK-004

Strict terminal binding conflicts with branch switching.
Medium / High
Require destination enrolled terminal; document operator flow.

- RISK-005

Historical reports change after configuration switch.
Low / High
Immutable transaction snapshots and report regression tests.

- RISK-006

Dirty working tree contaminates
implementation.
High / Medium
Use a clean dedicated worktree and baseline inventory.

- RISK-007

Phase ledger numbering conflicts with
location docs.
Medium / Medium
Reconcile ledger before Phase 355.

- RISK-008

Global selector becomes an unsafe
second switch path.
Medium / High
Selector dispatches only to governed Shift workflow.

Architecture decisions

ADR
Decision
Sections

- ADR-001

Global entry point, Shift-owned operating transition, separate viewing context.
PLAN-03, PLAN-06

- ADR-002

Shared catalog identity with location-specific operational state.
PLAN-03, PLAN-04, PLAN-07

- ADR-003

Backend-derived scope and atomic/idempotent transition.
PLAN-05, PLAN-08, PLAN-09

- ADR-004

Pilot and feature-flag rollout with additive migration only.
PLAN-07, PLAN-09

Relative effort and feasibility

Area
Effort
Reason
Discovery and contract
MEDIUM
Existing foundations reduce greenfield work; module scope is incomplete.
Backend enforcement
HIGH
Financial, inventory, authorization, and concurrency boundaries.
Frontend context and UX
MEDIUM
Existing state exists but needs central coordination and responsive UI.
Migration
UNKNOWN to HIGH
Depends on voucher, price-list, affiliate, and settings schema audit.
Testing and rollout

HIGH
Requires cross-module, security, migration, simulation, and recovery evidence.

PLAN-11 Final Recommendation and Execution Handoff

Final recommendation

Build a shared global location context with a visible header/sidebar entry point. Reuse the existing Shift-controlled operating-location transition. Keep viewing-location navigation read-only and separate from register operation. Extend only the
module contracts proven incomplete by Phase 354 discovery.

What must be built

- Global Current Location selector and explicit operating/viewing labels.
- Location context coordinator with stale-request protection and module refresh rules.
- Server-side location grants, module permissions, terminal binding, shift ownership, idempotency, and audit enforcement.
- Location-aware behavior for items, stock, Add Item, sales, parked sales, reports, shifts, settings, vouchers, price lists, affiliates,

queue, and audit.

- Verification matrix, migration rehearsal, pilot rollout, observability, and recovery evidence.

What must remain unchanged

- The existing Shift switch-location endpoint and governed close/open semantics.
- The shared catalog source of truth and location inventory ledger.
- Historical transaction location and audit history.
- Terminal pairing and strict device safety rules.
- Unrelated user changes in the current dirty working tree.

Plan readiness audit

Gate
Status
Reason
Objective and problem
PASS
Defined in PLAN-01 and PLAN-02.
Current state
PASS WITH BASELINE
CAUTION
Existing code and docs inspected; working tree is dirty.
Target architecture
PASS
Operating/viewing/queue contexts defined.
Gate
Status
Reason
Requirements and rules
PASS
FR, NFR, SEC, DATA, OPS, MIG, BR, and AC IDs defined.
Security and data ownership
PASS
Backend authority and source-of-truth model defined.

Migration
NOT READY
Voucher, price-list, affiliate, and settings schema audit pending.
Verification
READY TO EXECUTE LATER
Matrix defined; evidence does not exist yet.
Phase ledger
NOT READY
Ledger reconciliation required before Phase 355.
Overall plan
NOT EXECUTION READY
Phase 354 discovery must close blocking unknowns.

Execution handoff package

When PLAN STATUS becomes EXECUTION READY, the ExecutionPlan skill must read the entire package: this master index, all
plan sections, requirements, business rules, ADRs, risk register, verification matrix, migration and testing plans, acceptance
criteria, deployment and rollback plans, assumptions, and known risks.

First eligible implementation move

Complete Phase 354 discovery and contract reconciliation in a clean dedicated worktree. Do not implement the global selector or
database changes before the scope matrix, permission matrix, and module ownership audit are approved.

Continuous phase sequence

Use the existing governed sequence: Phase 354 contract/discovery, Phase 355 backend enforcement, Phase 356 POS UI/context
integration, and Phase 357 hardening and acceptance. Reconcile the main phase ledger before recording new entries.

Final decision

PLAN STATUS: NOT EXECUTION READY: The plan is complete as a planning artifact, but implementation must wait for
the documented discovery gates. This is intentional and protects inventory, financial history, authorization boundaries, and rollback safety.